IMAT3721 – Agent Based Modelling and Parallel Computing

Under the Guidance of Dr. Amudhavel Jayavel

Coursework – Agent Based Modelling and Multi-Threading

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TITLE

"An Agent-Based Model of Emergency Department Dynamics Under Capacity Constraints"

1. Abstract

Emergency departments operate as complex, capacity-constrained systems in which patient outcomes emerge from the interaction of individual deterioration dynamics, spatial access, and treatment availability. This project develops an agent-based model implemented in NetLogo Web to explore how emergency department performance varies under differing demand, capacity, and spatial configurations. Patients are modelled as autonomous agents whose health deteriorates continuously over time, while treatment occurs intermittently under probabilistic success and capacity constraints. A scenario-based experimental methodology is employed to evaluate baseline operation, surge conditions, infrastructure expansion, and system collapse. Results demonstrate non-linear threshold behaviour, where modest increases in demand or reductions in capacity can trigger abrupt increases in mortality, predominantly driven by delayed access to care rather than treatment failure. Although simplified, the model provides qualitative insight into the mechanisms through which congestion, spatial inefficiency, and resource limitations shape emergency department outcomes, while also illustrating the applicability of agent-based modelling and conceptual parallelism to complex healthcare systems.

2. Introduction

Emergency departments (EDs) operate under persistent pressure due to unpredictable patient arrivals, limited treatment capacity, and the time-sensitive nature of medical care. In such environments, even modest increases in demand or small inefficiencies in access can lead to congestion, prolonged waiting times, and adverse patient outcomes. Understanding how these factors interact at a system level is essential for designing resilient healthcare services, particularly in the context of increasing population pressures and constrained resources.

A defining characteristic of emergency care systems is that patient outcomes depend not only on the quality of treatment delivered, but also on the ability of patients to access care in time. Delays caused by congestion, spatial inefficiency, or limited staffing can result in patient deterioration before treatment begins, producing mortality that is indirectly driven by access failure rather than treatment failure. These dynamics are inherently non-linear: systems may appear stable under normal operating conditions yet experience rapid and disproportionate collapse once critical thresholds are exceeded.

Traditional analytical and aggregate modelling approaches often struggle to capture such behaviour, as they rely on averaged assumptions and equilibrium conditions. In contrast, agent-based modelling (ABM) provides a framework in which individual patients can be represented as autonomous agents whose health evolves over time and whose interactions with shared resources generate emergent system-level outcomes. By modelling patients individually rather than as homogeneous flows, ABM enables the study of congestion, competition for limited capacity, and threshold-driven regime shifts that arise from local interactions rather than central coordination.

From a computational perspective, agent-based models are naturally suited to conceptually parallel execution. Individual agents operate independently, updating their internal state and making decisions concurrently, while interacting through shared environmental constraints. In emergency department systems, treatment capacity represents a critical shared resource, introducing contention and access delays that strongly influence global behaviour. Although the present model is implemented in NetLogo Web, which executes agent updates synchronously rather than through explicit multi-threading, the system is deliberately structured around parallel design principles. This allows reasoning about scalability, contention, and shared-state dynamics within the constraints of the platform.

This project presents an agent-based model of emergency department operations in which patients experience continuous health deterioration, seek care once critical thresholds are reached, and compete for limited treatment capacity. The environment incorporates spatial structure, access pathways, and multiple emergency treatment areas, enabling investigation of both capacity-related and access-related failure modes. Treatment effects are applied intermittently to reflect realistic care delivery, while deterioration continues continuously, allowing congestion effects to emerge naturally.

The aim of this project is to investigate how individual patient deterioration, spatial access constraints, and finite treatment capacity interact to produce congestion, mortality, and systemic failure in emergency departments. Through a series of controlled experimental scenarios—including baseline operation, demand surges, resource limitations, preventive care, and extreme mass casualty events—the model seeks to identify non-linear threshold effects, emergent operational regimes, and the relative importance of access delays versus treatment effectiveness in determining patient outcomes.

By combining agent-based modelling with conceptually parallel system design, this study aims to provide insight into why emergency care systems often fail abruptly rather than gradually, and to demonstrate the value of ABM as a tool for analysing complex, capacity-constrained service environments.

This model is not intended as a predictive clinical decision-support tool. Instead, it is designed as an exploratory simulation that isolates core system-level mechanisms—such as access delay, capacity constraints, and spatial congestion—in order to study how emergency department outcomes emerge from local interactions under varying conditions.

3. Methodology and Model Design

This project employs an agent-based modelling (ABM) methodology to investigate how patient deterioration, access delays, and finite treatment capacity interact to produce congestion, mortality, and systemic failure in emergency department (ED) systems. Agent-based modelling is particularly well suited to this problem because it allows individual-level health dynamics, decentralised decision-making, and competition for shared resources to generate emergent, non-linear system behaviour.

The methodology consists of four stages: agent and environment design, mathematical formulation of health and movement dynamics, modelling of treatment and capacity constraints, and experimental scenario definition. Across all scenarios, patient decision rules and mathematical formulations remain fixed, ensuring that observed differences in outcomes arise from systemic constraints rather than behavioural changes.

3.1 Modelling Approach

Emergency departments are characterised by unpredictable patient arrivals, time-sensitive deterioration, and limited treatment resources. These features make ED systems difficult to analyse using aggregate or equilibrium-based approaches, which often obscure access delays and threshold effects.

In this model, patients are represented as autonomous agents whose health evolves over time. Agents interact indirectly through shared environmental constraints—most notably emergency department capacity—allowing congestion, delay, and failure to emerge naturally from local interactions rather than being imposed by design.

3.2 Patient Agent Design

Each patient is modelled as an individual agent representing a person seeking emergency medical care. Patient agents are defined by a small set of internal state variables governing health, movement, and treatment status.

3.2.1 Health Representation and Deterioration

Patient health is represented as a continuous scalar variable H. At each simulation tick, health deteriorates by a fixed amount representing the progression of untreated medical conditions.

Health update rule:

$H(t + 1) = \max(0, H(t) - d)$

where:

- $H(t)$ is the patient's health at time t
- d is the deterioration rate per tick

The lower bound of zero prevents negative health values and provides a clear death condition.

3.2.2 Critical Health Threshold

A critical health threshold H_c governs patient behaviour. When health falls below this threshold, patients actively seek emergency treatment. Above the threshold, patients move without urgency, representing delayed care-seeking or uncertainty.

Behavioural rule:

If $H < H_c \rightarrow$ patient seeks emergency care
If $H \geq H_c \rightarrow$ patient wanders

This threshold introduces non-linearity into the system, as small changes in deterioration or access delay can trigger abrupt behavioural transitions.

3.2.3 Movement Speed as a Function of Health

Patient movement speed is modelled as a non-linear function of health, creating feedback between deterioration and access efficiency.

Speed is calculated as:

speed = max(minSpeed, (sqrt(H) / 10) × speedFactor)

This formulation ensures that:

- movement slows as health deteriorates
- deterioration increases access delays
- patients remain mobile even at low health due to a minimum speed

This coupling allows congestion effects to amplify health decline without being explicitly programmed.

3.2.4 Treatment State

Each patient maintains a Boolean treatment state indicating whether treatment has been initiated. Once admitted to treatment, patients remain in that state until recovery or death, preventing oscillation between treated and untreated conditions.

3.3 Environment and Spatial Design

3.3.1 Spatial Layout

The simulation environment is represented as a two-dimensional grid containing:

- an entrance region where patients arrive,
- corridor pathways for enhanced movement speeds,
- one or more emergency treatment areas (ER rooms).

The spatial layout is deliberately simplified while preserving functional distinctions between access zones and treatment zones. This abstraction allows investigation of congestion, wayfinding inefficiency, and spatial delay without unnecessary architectural complexity.

3.3.2 Emergency Department Capacity

Each ER room has a fixed treatment capacity C, representing staffing and resource limitations. At any time, the number of patients receiving treatment in a given ER must satisfy:

currentLoad ≤ C

ER admission therefore constitutes a shared critical resource. Patients arriving when capacity is full must wait, continuing to deteriorate while competing for access.

3.4 Treatment and Health Recovery Modelling

Health dynamics in the model are intentionally simplified to prioritise interpretability over clinical realism. Deterioration is applied deterministically at each simulation tick, while recovery occurs probabilistically at discrete treatment intervals. This temporal asymmetry reflects the continuous nature of untreated decline versus the episodic nature of medical intervention and allows mortality and recovery outcomes to emerge dynamically rather than being imposed through deterministic thresholds.

3.4.1 Intermittent Treatment Logic

Treatment is applied intermittently rather than continuously. At fixed treatment intervals, each treated patient receives a probabilistic health increase.

Treatment rule (applied every treatmentInterval ticks):

If random < successRate →
$H(t + 1) = H(t) + \text{treatmentRate}$

Between treatment intervals, deterioration continues uninterrupted. This design choice prevents unrealistically rapid recovery under congested conditions and was validated experimentally (see Section 6.1).

3.4.2 Deterioration During Treatment

While receiving treatment, patients continue to deteriorate at a reduced rate, reflecting ongoing medical risk even under care.

Health update during treatment:

$H(t + 1) = H(t) - (d \times \text{erDeteriorationFactor})$

This allows deaths during treatment to emerge under sustained overload rather than being artificially suppressed.

3.4.3 Overload Risk Effects

To capture the impact of overcrowding and sustained ER pressure, an additional stochastic deterioration is applied with low probability:

If random < erOverloadRisk →
H(t + 1) = H(t) − d

This mechanism represents complications or delays arising from system overload.

3.4.4 Recovery and Death Conditions

Patients recover and exit the system when their health reaches or exceeds a target value:

If H ≥ targetHealth → patient recovers

Patients die when health reaches zero:

If H ≤ 0 → patient dies

Deaths are classified based on whether the patient was receiving treatment, allowing separation of access-related mortality from treatment-related mortality.

3.5 Movement and Access Behaviour

Patient movement is governed by local rules rather than global optimisation. Patients above the critical health threshold wander within the environment. Once health falls below the threshold, patients orient toward emergency treatment areas and attempt admission.

Movement outcomes depend on health-dependent speed, corridor availability, and spatial layout. Congestion and inefficient wayfinding therefore increase access delays indirectly, amplifying deterioration and mortality.

3.5.1 Corridor Movement

Patient movement is further constrained by corridor-based navigation rules that represent structured hospital pathways. Corridors act as preferred movement channels, enabling faster and more directed motion compared to unrestricted areas of the environment. When a patient attempts to move into a corridor patch, movement speed is increased relative to open space, reflecting guided access and reduced navigational uncertainty.

Conversely, when patients encounter walls or non-navigable areas, movement direction is altered stochastically to prevent penetration of inaccessible regions. This local redirection introduces minor delays and path inefficiency without requiring global pathfinding or shortest-path optimisation.

By combining health-dependent speed with corridor-constrained navigation, the model allows access delays to emerge naturally from spatial layout. Poor corridor alignment, longer travel distances, or inefficient wayfinding therefore increase time-to-treatment indirectly, amplifying health deterioration and congestion effects. This approach enables investigation of spatial efficiency and layout design as contributors to access-related failure without introducing computationally expensive routing algorithms.

3.6 Conceptual Parallelism and Shared-State Interaction

The model is structured around conceptually parallel execution. Each patient agent independently updates health, movement, and decision logic at every tick. Interaction occurs only through shared environmental state, primarily emergency department capacity.

ER admission represents a critical section in which multiple agents compete for limited access. In a truly multi-threaded implementation, this interaction would require explicit synchronisation mechanisms. In NetLogo Web's synchronous execution model, atomic updates at tick boundaries ensure consistency while preserving the conceptual structure of parallel agent interaction.

This design enables meaningful reasoning about contention, scalability, and shared-state dynamics within platform constraints.

3.7 Experimental Scenario Design

Ten experimental scenarios were defined to explore system behaviour under diverse conditions. Scenarios vary parameters including:

- patient demand,
- treatment capacity,
- deterioration rates,
- spatial efficiency,
- preventive intervention effects.

Importantly, agent behaviour and mathematical rules remain identical across all scenarios, ensuring that differences in outcomes reflect systemic constraints rather than behavioural changes.

3.8 Simulation Execution and Data Collection

Each scenario was executed multiple times to account for stochastic variation arising from probabilistic treatment success and movement behaviour. Aggregate outcome measures—including survival counts, mortality classification, ER load, and system clearance time—were recorded alongside time-series data capturing ER pressure, patient outcomes, and health dynamics.

These outputs form the basis for the comparative analysis presented in Section 6.

3.9 Interface Elements as Experimental Controls

The model interface provides a set of adjustable parameters that function as experimental controls rather than user-driven gameplay elements. Sliders, switches, and monitors are used to systematically vary model conditions across scenarios while holding agent behaviour and mathematical rules constant.

Key interface controls include parameters governing patient arrival pressure, deterioration rate, treatment capacity, treatment success probability, spatial layout configuration, and preventive intervention effects. By adjusting these parameters between simulation runs, distinct experimental scenarios are defined without modifying underlying agent logic or interaction rules.

This design ensures reproducibility and comparability across scenarios. All experimental variation is introduced exclusively through controlled parameter changes at the interface level, allowing causal relationships between system constraints and emergent outcomes to be isolated and analysed.

Screenshots of the interface and parameter configurations for each scenario are provided to document experimental conditions and support reproducibility.

3.10 Monitors, Plots, and Output Measurement

The interface includes a set of monitors and plots used to capture both aggregate outcomes and time-dependent system behaviour during simulation execution. These outputs serve as observational instruments rather than control mechanisms.

Monitors record cumulative values such as the number of patients saved, deaths occurring outside the emergency department, deaths occurring during treatment, and total simulation duration. These metrics form the basis of the quantitative comparisons presented in Section 6.

Time-series plots capture dynamic system behaviour, including emergency department load, patient outcome trajectories, and average health over time. These plots enable identification of congestion onset, recovery phases, and collapse regimes that are not evident from final outcome counts alone.

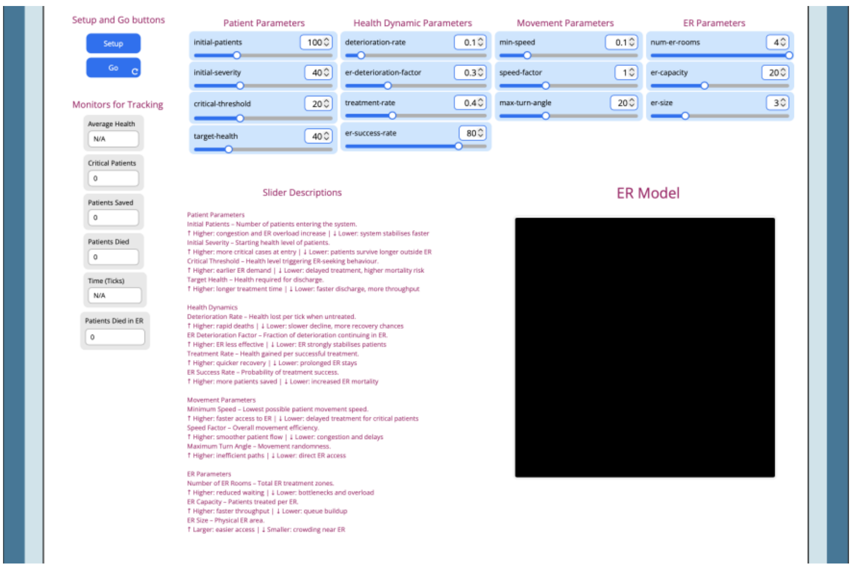
By separating interface controls (used to define experimental conditions) from interface outputs (used to observe system behaviour), the model maintains a clear distinction between manipulation and measurement, consistent with experimental modelling practice.

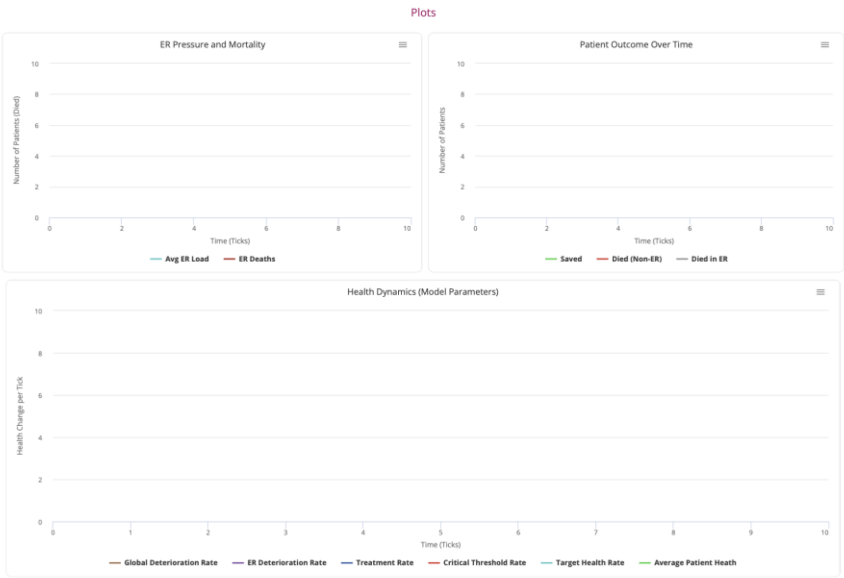
4. Implementation Details

All user-adjustable interface elements are restricted to parameters representing environmental conditions or system capacity. Structural assumptions, such as treatment interval timing and execution order, are deliberately implemented as internal model constants to prevent unrealistic optimisation through interface manipulation.

4.1 Interface Elements and Experimental Controls

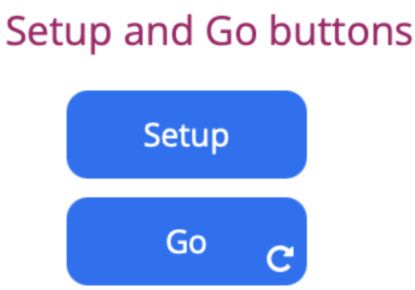
The NetLogo interface serves as the experimental control layer of the model. Each button, slider, monitor, and plot corresponds directly to specific variables and procedures in the underlying code, allowing systematic manipulation of parameters and transparent observation of emergent outcomes. This section documents each interface component explicitly and links it to its computational role.





NetLogo Web interface showing experimental control sliders, execution buttons, monitors, and output plots used to configure and observe emergency department simulations.

4.1.1 Execution Controls



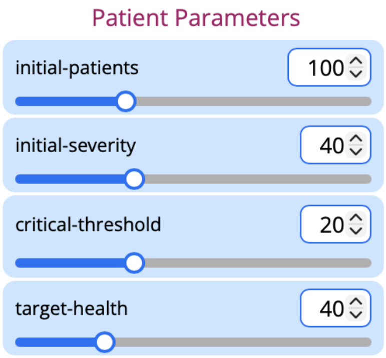
Setup Button

The Setup button calls the setup procedure, which clears the simulation state, resets all global counters, reconstructs the hospital environment, creates ER rooms, and generates a new patient population based on the current slider values. This ensures that each experimental run begins from a controlled and reproducible initial condition.

Go Button

The Go button repeatedly executes the go procedure, advancing the simulation by one tick per iteration. During each tick, patients deteriorate, move through the environment, seek emergency care when critical, and exit the system through recovery or death. Separating setup from execution allows repeated trials under identical parameter configurations.

4.1.2 Patient Parameter Sliders



Initial Patients (initial-patients)

Controls the number of patient agents created at setup. This parameter directly determines system load and congestion. Higher values increase competition for ER capacity and raise the likelihood of untreated deterioration.

Initial Severity (initial-severity)

Sets the baseline health level assigned to patients at creation, with small random variation. Lower values represent a sicker population at entry and increase the probability of early criticality and mortality.

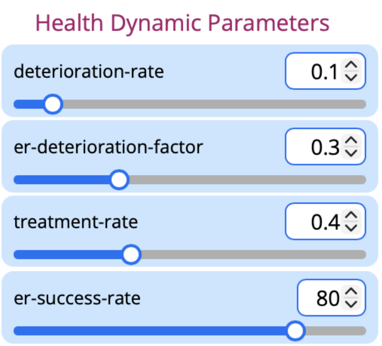
Critical Threshold (critical-threshold)

Defines the health level below which patients switch from random wandering to ER-seeking behaviour. This parameter directly affects the timing of ER demand and the spatial flow of agents toward treatment areas.

Target Health (target-health)

Specifies the health level at which a treated patient is considered recovered and discharged. Higher target values lengthen ER stays, reducing throughput and increasing congestion.

4.1.3 Health Dynamics Sliders



Deterioration Rate (deterioration-rate)

Represents the amount of health lost by each patient per tick when untreated. This value is applied every simulation tick via the deteriorate procedure, modelling continuous health decline.

ER Deterioration Factor (er-deterioration-factor)

Scales the deterioration rate while patients are receiving treatment. Values below one represent partial stabilisation in the ER, while values closer to one indicate ineffective treatment under overload.

Treatment Rate (treatment-rate)

Defines the amount of health gained during a successful treatment event. Treatment does not occur every tick; instead, it is attempted at fixed intervals, reflecting episodic medical intervention.

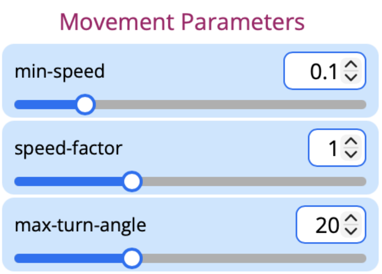
ER Success Rate (er-success-rate)

Represents the probability that a treatment attempt is successful when it occurs. This introduces stochasticity into recovery outcomes, reflecting variability in clinical effectiveness.

Important structural assumption

Treatment attempts occur every three ticks, controlled by an internal treatment-interval parameter. In contrast, deterioration occurs every tick. This temporal asymmetry models the reality that health declines continuously, while treatment is intermittent.

4.1.4 Movement Parameter Sliders



Minimum Speed (min-speed)

Sets the lower bound on patient movement speed. Patients with very low health cannot move arbitrarily slowly, preventing unrealistic immobility.

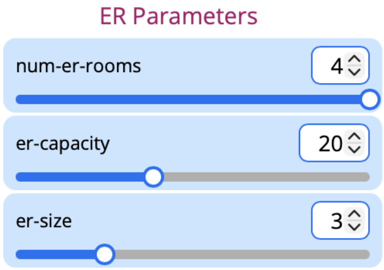
Speed Factor (speed-factor)

Scales movement speed as a function of patient health. Healthier patients move more efficiently, while sicker patients move more slowly, indirectly increasing mortality risk.

Maximum Turn Angle (max-turn-angle)

Controls directional randomness during wandering. Higher values produce less efficient paths to the ER, increasing delays and congestion in corridors.

4.1.5 Emergency Department Parameter Sliders



Number of ER Rooms (num-er-rooms)

Controls how many distinct ER treatment zones are created in the environment. Increasing this parameter spatially distributes treatment capacity and reduces congestion effects.

ER Capacity (er-capacity)

Defines the maximum number of patients that can be treated simultaneously in each ER room. Once capacity is reached, additional patients cannot receive treatment even if physically present.

ER Size (er-size)

Controls the physical footprint of each ER room. Larger ERs allow more patients to physically occupy the treatment area, indirectly affecting queueing behaviour.

4.1.6 System Monitors

Patients Saved

Counts the total number of patients discharged after reaching the target health threshold.

Patients Died

Tracks deaths occurring outside the ER due to untreated deterioration.

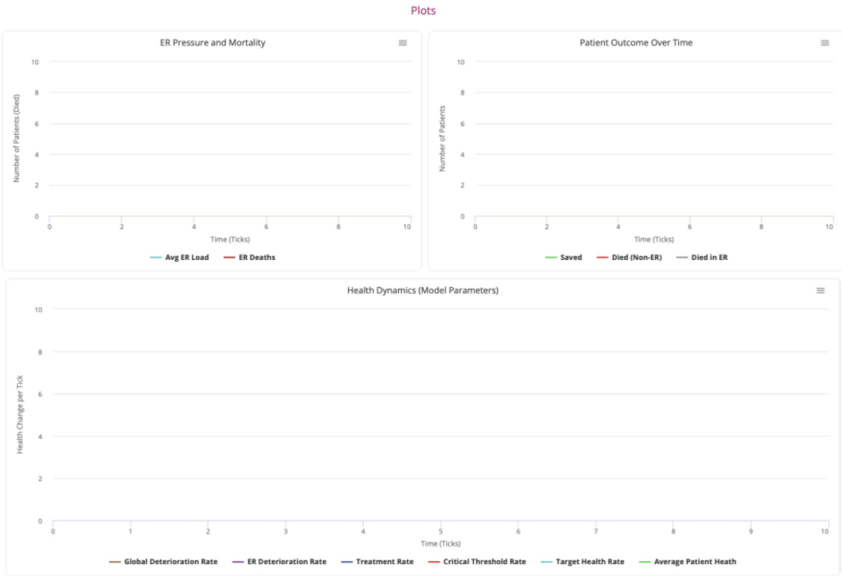
Patients Died in ER

Tracks deaths occurring while patients are undergoing treatment, capturing failures due to overload or ineffective care.

Average Health / Critical Patients / Time (Ticks)

Provide real-time indicators of system state, allowing identification of congestion points and collapse dynamics during execution.

4.1.7 Dynamic Plots



ER Pressure and Mortality Plot

Visualises the relationship between average ER load and deaths over time, highlighting overload-driven mortality dynamics.

Patient Outcome Over Time Plot

Separately tracks recovered patients, non-ER deaths, and ER deaths, enabling distinction between access-related and treatment-related failure modes.

Health Dynamics Plot

Displays deterioration rates, treatment effects, and average patient health, providing visual validation that health trajectories evolve according to the model's equations.

4.1.8 Role of the Interface in Experimental Validity

The interface is intentionally designed to expose parameters that define experimental scenarios while fixing structural assumptions such as treatment interval timing. This prevents unrealistic optimisation through interface tuning and ensures that observed outcomes emerge from agent interactions and system constraints rather than manual intervention.

4.2 Overview of Code Structure

The model is implemented in NetLogo Web and is structured to clearly separate environment setup, agent initialisation, state update logic, and outcome tracking. This modular organisation improves readability, supports debugging, and ensures that individual components of the system can be modified independently during scenario experimentation.

At a high level, the implementation consists of:

- global variables to track system-level outcomes and ER loads
- patient agents with internal state variables
- patch-based environment logic defining hospital zones
- a main simulation loop governing system evolution
- specialised procedures for health dynamics, movement, and treatment

This separation mirrors the conceptual model described in the previous section and ensures a direct correspondence between design and implementation.

4.3 Global State and System Metrics

```
globals [
  patients-saved
  patients-died
  patients-died-in-er
  er-room-loads
  empty-tick
  treatment-interval
  er-overload-risk
]
```

The model defines a set of global variables to track aggregate system outcomes and shared state that cannot be attributed to individual agents alone. These variables enable the measurement of system performance, capacity pressure, and failure modes across simulation runs and experimental scenarios.

patients-saved records the cumulative number of patients who successfully recover after receiving emergency department treatment.

patients-died counts deaths occurring before entry into emergency care, capturing access-related failure modes.

patients-died-in-er tracks deaths occurring during treatment, allowing separation of access-related and treatment-related mortality.

er-room-loads is implemented as a list, with each index corresponding to the occupancy of a specific ER room.

empty-tick records the time at which the final patient exits the system, supporting controlled termination logic.

treatment-interval defines the number of ticks between treatment attempts and is implemented as an internal model constant.

er-overload-risk represents the probability of acute deterioration during ER treatment under sustained congestion.

Together, these variables capture both aggregate performance and spatially localised failure modes while maintaining a clear distinction between experimental controls and internal mechanics.

4.4 Environment Setup and Spatial Encoding

4.4.1 Boundary and Layout Construction

```
to setup-boundary
  ask patches [
    if pxcor = max-pxcor or pxcor = min-pxcor or
       pycor = max-pycor or pycor = min-pycor [
      set pcolor white
    ]
    if pcolor != white [
      ifelse (pxcor + pycor) mod 2 = 0
      [ set pcolor 9 ]
      [ set pcolor 8 ]
    ]
  ]
end
```

```
to setup-corridors
  ask patches with [ abs pxcor <= 1 and abs pycor <= 14 ] [ set pcolor 6 ]
  ask patches with [ abs pycor <= 1 and abs pxcor <= 14 ] [ set pcolor 6 ]
end
```

The hospital environment is constructed using patch-level procedures that assign colours and attributes to represent walls, floors, corridors, entry zones, and ER rooms. Boundary patches prevent agents from exiting the system, ensuring a closed simulation environment.

Corridors are implemented as preferred movement pathways, visually distinct and associated with increased movement efficiency. This encoding allows spatial structure to influence agent behaviour without requiring explicit pathfinding algorithms.

4.4.2 Entry and ER Zones

```
to setup-entrance
  ask patches with [
    pxcor > -3 and pxcor < 3 and
    pycor > -3 and pycor < 3
  ] [
    set pcolor 7
  ]
end
```

```
to setup-er-rooms
  set er-room-loads n-values num-er-rooms [0]

  ask patches with [pcolor = green] [
    set pcolor black
    set er-id nobody
  ]

  let s er-size

  if num-er-rooms >= 1 [
    ask patches with [
      pxcor >= (-14) and pxcor < (-14 + 2 * s) and
      pycor <= (14) and pycor > (14 - 2 * s)
    ] [
      set pcolor green
      set er-id 0
    ]
  ]

  if num-er-rooms >= 2 [
    ask patches with [
      pxcor <= (14) and pxcor > (14 - 2 * s) and
      pycor <= (14) and pycor > (14 - 2 * s)
    ] [
      set pcolor green
      set er-id 1
    ]
  ]

  if num-er-rooms >= 3 [
    ask patches with [
      pxcor >= (-14) and pxcor < (-14 + 2 * s) and
      pycor >= (-14) and pycor < (-14 + 2 * s)
    ] [
      set pcolor green
      set er-id 2
    ]
  ]

  if num-er-rooms >= 4 [
    ask patches with [
      pxcor <= (14) and pxcor > (14 - 2 * s) and
      pycor >= (-14) and pycor < (-14 + 2 * s)
    ] [
      set pcolor green
      set er-id 3
    ]
  ]
end
```

The entry zone is defined as a central triage area where all patients are initially spawned. ER rooms are implemented as spatially distinct rectangular regions, each assigned a unique identifier and tracked independently through the er-room-loads list.

The number and size of ER rooms are parameterised, allowing the same codebase to represent hospitals with varying infrastructure capacities.

4.5 Patient Initialisation

```
to setup-patients
  create-patients initial-patients [
    move-to one-of entrance-patches
    set size 1.2
    set shape "person"

    set health max list 1 (initial-severity + random 3 - random 3)
    set speed max list min-speed ((sqrt health / 10) * speed-factor)

    set being-treated? false
    update-colour
    set label precision health 0
    set label-color white
  ]
end
```

Patients are created using the create-patients procedure and placed randomly within the entry zone. Each patient is initialised with:

- a health value derived from the initial-severity parameter with bounded randomness
- a speed value computed as a function of health
- a treatment status flag initialised to false

Health labels and colour coding are applied for visual clarity but do not influence simulation logic.

4.6 Main Simulation Loop

```
to go
  if empty-tick != -1 and ticks >= empty-tick + 10 [ stop ]

  if (not any? patients) and empty-tick = -1 [
    set empty-tick ticks
  ]

  ask patients [

    ;; ER patients handled first
    if being-treated? [
      receive-treatment
      update-colour
      set label precision health 0
      stop
    ]

    ;; death outside ER
    if health <= 0 [
      set patients-died patients-died + 1
      die
    ]

    deteriorate
    decide-movement

    set speed max list min-speed ((sqrt health / 10) * speed-factor)
    update-colour
    set label precision health 0
  ]

  tick
end
```

The simulation progresses through the go procedure, representing a single discrete time step. Each tick consists of a synchronous update of all patient agents, reflecting conceptual parallelism. Termination is handled using a delayed stopping condition. Once the system becomes empty, the simulation continues for a fixed number of additional ticks before halting, preventing premature termination during transient clearance states.

The simulation continues until all patient agents have exited the system through either recovery or death, after which execution is delayed briefly before termination to ensure consistent observation of final system states.

4.7 Agent-Level State Updates

4.7.1 Execution Order and Concurrency

Within each tick, all patients are updated using an ask patients construct. Conceptually, this represents concurrent execution. Although NetLogo Web executes updates sequentially, the model is structured to avoid order-dependent artefacts, aside from shared constraints such as ER capacity.

4.7.2 Health Deterioration and Non-ER Deaths

```
to deteriorate
  set health health - deterioration-rate
  if health < 0 [ set health 0 ]
end
```

Patients not receiving treatment experience health deterioration at every tick. If health reaches zero outside the ER, the patient is immediately removed and counted as a non-ER death.

An intentional temporal asymmetry is introduced: deterioration occurs every tick, while treatment effects are applied intermittently. This asymmetry reflects clinical reality and is validated empirically in Section 6.

4.7.3 Treatment Logic and ER Deaths

```
to receive-treatment
  if not member? patch-here er-patches [ stop ]
  let id [er-id] of patch-here

  ;; intermittent treatment
  if ticks mod treatment-interval = 0 [
    if random-float 100 < er-success-rate [
      set health health + treatment-rate
    ]
  ]

  ;; deterioration continues
  let er-deterioration deterioration-rate * er-deterioration-factor
  set health health - er-deterioration

  ;; acute overload shock
  if random-float 100 < er-overload-risk [
    set health health - deterioration-rate
  ]

  ;; ER death
  if health <= 0 [
    set patients-died-in-er patients-died-in-er + 1
    set er-room-loads replace-item id er-room-loads
      max list 0 (item id er-room-loads - 1)
    die
  ]

  ;; recovery
  if health >= target-health [
    set patients-saved patients-saved + 1
    set er-room-loads replace-item id er-room-loads
      max list 0 (item id er-room-loads - 1)
    die
  ]
end
```

Patients may begin treatment only if capacity is available and recovery has not already occurred. Treatment effects are applied probabilistically at fixed intervals, while deterioration continues at a reduced rate. An additional stochastic overload risk allows deaths during treatment to emerge under sustained congestion.

If health reaches zero during treatment, the event is recorded as an ER death and capacity counters are updated accordingly. If health reaches the recovery threshold, the patient is discharged successfully. Earlier continuous-treatment variants were retained primarily for validation and sensitivity analysis and are not used in final scenario evaluation.

4.8 Movement and Navigation Logic

```
to decide-movement
  if health < critical-threshold [ move-to-er ]
  if health >= critical-threshold [ wander ]
end

to wander
  if speed > 0.01 [
    let turn-angle (max-turn-angle * (health / 100))
    rt random-float turn-angle - random-float turn-angle
    attempt-move
  ]
end

to move-to-er
  let target one-of er-patches

  if target != nobody [
    face target
    attempt-move
  ]

  if member? patch-here er-patches [
    let id [er-id] of patch-here
    let current-load item id er-room-loads

    if (not being-treated?) and current-load < er-capacity and health < target-health [
      set being-treated? true
      set er-room-loads replace-item id er-room-loads (current-load + 1)
    ]
  ]
end

to attempt-move
  if speed < 0.01 [ stop ]

  let next-patch patch-ahead speed

  if next-patch != nobody and [pcolor] of next-patch = white [
    set heading heading + 180 + random 60 - random 60
    fd speed * 0.5
    stop
  ]

  if next-patch != nobody and [pcolor] of next-patch = 6 [
    fd speed * 1.2
  ]
  if next-patch = nobody or [pcolor] of next-patch != 6 [
    fd speed
  ]
end
```

Patient movement distinguishes between routine mobility and emergency care-seeking. Above the critical threshold, agents wander with stochastic turning and corridor bias. Below the threshold, agents attempt to seek emergency care but do not assume perfect spatial knowledge. Emergency room targets are selected stochastically rather than deterministically, representing stress and limited wayfinding. Navigation inefficiencies, congestion, and delayed access therefore emerge organically without enforcing optimal routing behaviour.

4.9 Error Handling and Stability Measures

```
  if health <= 0 [
    set patients-died patients-died + 1
    die
  ]
```

```
if health <= 0 [
  set patients-died-in-er patients-died-in-er + 1
  set er-room-loads replace-item id er-room-loads
    max list 0 (item id er-room-loads - 1)
  die
]
```

```
  if health >= target-health [
    set patients-saved patients-saved + 1
    set er-room-loads replace-item id er-room-loads
      max list 0 (item id er-room-loads - 1)
    die
  ]
end
```

```
if empty-tick != -1 and ticks >= empty-tick + 10 [ stop ]

if (not any? patients) and empty-tick = -1 [
  set empty-tick ticks
]
```

Safeguards are implemented to ensure numerical and logical stability:

- health values are bounded
- ER loads are clamped to non-negative values
- capacity checks prevent over-allocation
- agents are removed immediately upon death or recovery

These measures ensure consistent behaviour across long and repeated runs.

4.10 Conceptual Parallelism and Scalability Considerations

Although NetLogo Web does not support explicit multi-threading, the model is designed around agent independence. Each patient update involves only local state and shared read-only environment data, with limited contention over ER capacity.

In lower-level implementations, patient updates could be parallelised with synchronisation required only for ER admission events, demonstrating that the model structure is inherently compatible with true parallel execution.

4.11 Implementation Challenges

Challenges included accurate ER load tracking, preventing post-recovery treatment, balancing deterioration and treatment rates, and managing termination conditions. These were addressed through incremental testing and explicit state checks.

4.12 Summary

The implementation faithfully translates the conceptual model into a stable and extensible simulation. Modular code structure, agent-level concurrency, and explicit capacity constraints support robust comparative analysis across diverse emergency healthcare scenarios.

5. Experimental Design and Scenario Configuration

A scenario-based experimental methodology is used to evaluate system behaviour under varying demand, capacity, spatial configuration, and treatment effectiveness. Ten scenarios are defined to represent plausible real-world healthcare conditions, including baseline operation, seasonal surges, resource shortages, infrastructure expansion, preventive intervention, and worst-case system collapse.

Each scenario is executed three times using identical parameter configurations to account for stochastic variation arising from probabilistic treatment outcomes and randomised agent movement. Aggregate metrics are averaged across runs to identify consistent system-level trends rather than run-specific noise.

5.1 Experimental Methodology

The primary objective of the experimental phase of this project is to evaluate how variations in patient demand, treatment capacity, spatial design, and health dynamics influence emergency department outcomes. Given the stochastic nature of the model and the non-linear interactions between agents, a scenario-based experimental design was adopted.

Rather than attempting exhaustive exploration of the parameter space, the model is evaluated using a set of carefully constructed scenarios, each representing a distinct and plausible emergency healthcare condition. This approach enables meaningful comparison between qualitatively different operating regimes while maintaining interpretability of results.

Scenario differences are implemented exclusively through controlled changes to interface parameters, while agent decision rules, health update logic, and treatment timing mechanisms remain fixed. Each simulation run is initialised using the same setup procedure, ensuring reproducibility and comparability across experiments.

Each scenario is executed multiple times using identical parameter configurations. Three replications were selected as a balance between computational feasibility and the need to identify consistent qualitative trends, rather than to support formal statistical inference.

5.2 Rationale for Scenario-Based Analysis

Emergency departments rarely operate under a single, stable set of conditions. Instead, they experience fluctuating patient loads, changing resource availability, and occasional extreme shocks. Scenario-based analysis provides a structured framework for examining how such variations affect system performance and patient outcomes.

This approach is particularly well suited to agent-based modelling, as it allows emergent behaviour to be observed under contrasting conditions without altering underlying agent logic. By holding agent decision rules constant across all scenarios and modifying only environmental and systemic parameters, the analysis isolates the causal impact of demand intensity, resource capacity, spatial configuration, and treatment effectiveness on emergent system behaviour.

The selected scenarios are designed to span multiple operational regimes, including stable throughput, prolonged congestion, and system collapse, allowing threshold effects and non-linear transitions to be identified.

5.3 Overview of Experimental Metrics

Across all scenarios, the following key metrics are recorded:

- **Total simulation time (ticks):** indicating how long the system remains active before all patients either recover or die, serving as a proxy for system clearance efficiency.

- **Number of patients saved:** representing successful treatment outcomes and overall system effectiveness.
- **Number of deaths outside the ER:** capturing access-related failure modes, including delayed arrival, congestion, and inability to reach treatment in time.
- **Number of deaths within the ER:** indicating treatment failure under capacity constraints and clinical overload conditions.
- **Mortality percentage:** a normalised failure metric enabling comparison across scenarios with differing patient volumes and demand intensities.

Together, these metrics provide complementary perspectives on emergency department performance and allow distinction between access-driven mortality, treatment-driven mortality, and system-wide failure under stress.

5.4 Scenario Definitions

Ten scenarios were designed to evaluate emergency department performance across baseline, stress, intervention, and collapse regimes. In each scenario, agent decision logic remains unchanged; only system-level parameters (demand, capacity, spatial configuration, and health dynamics) are modified.

All scenarios are evaluated under two treatment implementations:

- **Table 1** reports results from the baseline (continuous-treatment) configuration, used primarily for validation and comparative reference.
- **Table 2** reports results from the final **intermittent-treatment model**, which incorporates temporally discrete treatment and overload risk and forms the basis for all substantive analysis.

Unless otherwise stated, scenario interpretations in this section refer primarily to **Table 2**, with Table 1 used to highlight contrasts in system behaviour under alternative treatment assumptions.

Scenario 1: Baseline Emergency Department

This scenario represents a moderately resourced emergency department operating under typical weekday conditions. Patient demand, ER capacity, treatment effectiveness, and deterioration rates are balanced to produce stable system behaviour.

Across both tables, this scenario exhibits low mortality and moderate clearance times, establishing a reference baseline against which all other scenarios are compared.

Scenario 2: Seasonal Illness Surge

This scenario simulates a seasonal increase in patient demand, such as during a winter influenza surge. Initial patient numbers are substantially increased while ER capacity and treatment parameters remain unchanged.

Results in both tables demonstrate increased congestion, longer clearance times, and sharply higher mortality, illustrating the system's sensitivity to demand shocks even in the absence of infrastructure or staffing changes.

Scenario 3: Resource-Limited / Understaffed Hospital

This scenario represents an understaffed or resource-constrained emergency department. ER capacity and treatment effectiveness are reduced while patient demand remains close to baseline levels.

Both tables show severe access-related failure, with very low numbers of patients successfully treated and mortality approaching system collapse levels in the intermittent-treatment model (Table 2). This scenario isolates the impact of reduced treatment availability independent of demand surges.

Scenario 4: Well-Funded Urban Trauma Centre

This scenario models a well-resourced urban trauma centre. ER capacity is increased, treatment effectiveness is improved, and deterioration is modestly reduced to reflect higher staffing levels, specialist equipment, and improved clinical workflows.

Results show improved survival and reduced congestion compared to baseline, though gains are non-linear. The comparison between Tables 1 and 2 demonstrates that increased resources mitigate but do not eliminate mortality under stochastic and intermittent treatment dynamics.

Scenario 5: Delayed Care / Low Health Awareness

This scenario represents delayed presentation due to low health awareness or poor access to early care. Patients enter the system with worse initial health, increasing the likelihood of rapid deterioration before treatment can begin.

Both tables show increased mortality relative to baseline despite unchanged capacity, highlighting the dominant role of early access and severity at entry in determining outcomes.

Scenario 6: Mass Casualty Event

This scenario simulates a sudden mass casualty event, such as a major accident or disaster. Patient demand is increased dramatically while ER capacity remains limited.

Results in both tables show extreme congestion, prolonged system activity, and near-total mortality in the intermittent-treatment model. This scenario exposes tipping points beyond which emergency department function collapses rapidly.

Scenario 7: ER Infrastructure Expansion

This scenario represents long-term infrastructure investment through the addition of ER rooms. Spatial treatment capacity is expanded while patient demand remains unchanged.

Outcomes show improved throughput and reduced mortality relative to baseline, though benefits are constrained by treatment rates and access dynamics. The scenario demonstrates that spatial expansion alone alleviates but does not fully resolve congestion under stochastic treatment.

Scenario 8: Poor Hospital Layout and Wayfinding

This scenario models poor hospital layout or inadequate wayfinding. Corridor efficiency and movement speed are reduced, increasing travel time to emergency care despite unchanged demand and capacity.

Results show elevated mortality driven primarily by deaths occurring outside the ER, reinforcing the importance of spatial design and navigation efficiency in access-related failure.

Scenario 9: Preventive Care and Early Intervention

This scenario represents the effects of strong preventive care and early intervention. Patients enter the system with higher initial health and slower deterioration, reflecting earlier diagnosis and effective triage.

Both tables show significantly reduced mortality and faster system clearance, demonstrating that upstream interventions can substantially reduce downstream emergency department pressure even without increasing capacity.

Scenario 10: Worst-Case System Failure

This scenario represents extreme system collapse under pandemic-level conditions. Patient demand is high, ER capacity is minimal, and treatment effectiveness is reduced.

Both tables show complete or near-complete mortality and prolonged system activity, illustrating dominant failure modes when multiple stressors interact. This scenario defines the upper bound of system instability and serves as a contrast to all intervention scenarios.

5.5 Replication and Robustness Considerations

Each scenario is executed three times using identical parameter configurations. While full statistical analysis is beyond the scope of this project, replication reduces the influence of stochastic variation and supports identification of consistent qualitative trends.

Observed variations between runs are analysed qualitatively and incorporated into interpretation, particularly for scenarios exhibiting chaotic behaviour or near-collapse dynamics.

5.6 Summary

The experimental design employs a structured, scenario-based methodology to explore how emergency department outcomes emerge under varying operational conditions. By maintaining consistent agent behaviour across all scenarios and introducing variation only through controlled parameter changes, the analysis isolates the effects of demand, capacity, spatial design, and treatment effectiveness. This framework provides a robust foundation for the comparative results presented in the following section.

6. Results and Comparative Analysis

This section presents the results of the agent-based simulation across ten experimental scenarios representing varying levels of demand, capacity, spatial efficiency, and treatment effectiveness. Results are analysed using both aggregate outcome metrics and time-series visualisations capturing ER load, patient outcomes, and population-level health dynamics.

The analysis proceeds in three stages. First, an intermediate validation is presented to justify the treatment timing assumptions used in the final model. Second, aggregate quantitative outcomes across all scenarios are compared using summary tables. Finally, scenario-wise time-series plots are interpreted to identify emergent behavioural regimes, threshold effects, and dominant failure modes.

Because all scenarios share identical agent decision rules, observed differences in outcomes arise directly from environmental constraints and parameter configurations rather than behavioural reprogramming. This ensures that comparative results reflect genuine system-level dynamics rather than artefacts of rule changes.

Observed differences across scenarios arise solely from changes in system-level constraints, as agent behaviour and decision rules are held constant throughout all experiments. This ensures that variations in outcomes reflect emergent system dynamics rather than behavioural reprogramming.

6.1 Validation of Treatment Timing Assumptions

Before presenting final scenario outcomes, an intermediate validation analysis was conducted to assess the impact of treatment timing on emergent system behaviour. Two treatment regimes were compared:
(i) **continuous treatment**, in which positive treatment effects were applied at every simulation tick, and
(ii) **intermittent treatment**, in which treatment was attempted at fixed temporal intervals while deterioration continued at every tick.

Results from the continuous-treatment implementation are presented in **Table 1** (**continuous treatment**) and are retained solely for methodological comparison. Under continuous treatment, mortality remained artificially suppressed across multiple scenarios, including those characterised by high patient load and limited capacity. Emergency departments appeared unrealistically stabilising, with rapid recovery even under conditions of sustained congestion.

Introducing intermittent treatment produced qualitatively different outcomes. As shown in **Table 2 (intermittent treatment)**, mortality increased consistently under stressed scenarios, and system failure emerged naturally when access delays and capacity constraints compounded deterioration. These results confirm that treatment timing is a critical determinant of emergent dynamics and justify the use of discrete treatment intervals in the final model. All subsequent analysis is therefore based exclusively on the intermittent-treatment implementation.

Table 1 with exploded view of tables given below for visibility (continuous treatment)

Scenario	Real World Analogy	Experimental Parameter Configuration													
		IP	IS	CT	TH	DR	ERDF	TR	ERSR	MS	SF	MTA	NERR	ERC	ERS
Normal Day Baseline ER	Typical weekday public hospital	80	35	20	40	0.08	0.25	0.45	85	0.1	1	20	3	20	3
Seasonal Flu Surge	Winter influenza increase	180	45	20	45	0.15	0.3	0.4	75	0.1	1	20	4	20	3
Resource Limited / Understaffed Hospital	Rural or low-income ER	120	45	30	35	0.14	0.5	0.25	60	0.07	0.8	20	1	10	2
Well Funded Urban Trauma Centre	Major metropolitan trauma hospital	140	50	15	50	0.07	0.2	0.6	90	0.12	1.2	25	4	30	4
Delayed Care / Low Health Awareness	Late presentation due to poor awareness	100	55	35	45	0.18	0.4	0.4	75	0.06	0.9	20	4	20	3
Mass Casualty Event	Accident / disaster influx	300	65	10	30	0.22	0.5	0.3	65	0.15	1.3	30	4	20	3
ER Infrastructure Expansion	New ER wing added	200	45	20	45	0.12	0.3	0.45	85	0.11	1	20	4	35	4
Poor Hospital Layout / Wayfinding	Confusing corridors & signage	120	40	20	40	0.12	0.3	0.4	80	0.05	0.7	40	4	20	3
Preventive Care & Early Intervention	Strong primary healthcare system	60	30	15	50	0.06	0.2	0.5	90	0.12	1	20	4	20	3
Worst Case System Failure	Pandemic-level collapse	260	70	40	55	0.25	0.6	0.2	50	0.05	0.8	25	1	10	2

Scenario	Results														
	Time (Ticks)			Patients Saved			Patients Died (Outside ER)			Patients Died (In ER)			Mortality Percentage		
	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3
Normal Day Baseline ER	474	493	520	70	66	64	10	14	16	0	0	0	12.5	17.5	20
Seasonal Flu Surge	481	467	481	98	97	97	82	83	83	0	0	0	45.56	46.11	46.11
Resource Limited / Understaffed Hospital	758	756	721	22	23	21	98	97	99	0	0	0	81.67	80.83	82.5
Well Funded Urban Trauma Centre	810	761	759	105	109	112	35	31	28	0	0	0	25	22.14	20
Delayed Care / Low Health Awareness	503	426	443	91	84	85	9	16	15	0	0	0	9	16	15
Mass Casualty Event	589	608	571	80	80	80	220	220	220	0	0	0	73.33	73.33	73.33
ER Infrastructure Expansion	481	507	482	142	147	153	58	53	47	0	0	0	29	26.5	23.5
Poor Hospital Layout / Wayfinding	396	472	432	75	77	64	45	43	56	0	0	0	37.5	35.83	46.67
Preventive Care & Early Intervention	545	574	528	48	51	56	12	9	4	0	0	0	20	15	6.67
Worst Case System Failure	1047	1016	945	0	0	0	250	250	250	10	10	10	100	100	100

Scenario	Real World Analogy	Experimental Parameter Configuration													
		IP	IS	CT	TH	DR	ERDF	TR	ERSR	MS	SF	MTA	NERR	ERC	ERS
Normal Day Baseline ER	Typical weekday public hospital	80	35	20	40	0.08	0.25	0.45	85	0.1	1	20	3	20	3
Seasonal Flu Surge	Winter influenza increase	180	45	20	45	0.15	0.3	0.4	75	0.1	1	20	4	20	3
Resource Limited / Understaffed Hospital	Rural or low-income ER	120	45	30	35	0.14	0.5	0.25	60	0.07	0.8	20	1	10	2
Well Funded Urban Trauma Centre	Major metropolitan trauma hospital	140	50	15	50	0.07	0.2	0.6	90	0.12	1.2	25	4	30	4
Delayed Care / Low Health Awareness	Late presentation due to poor awareness	100	55	35	45	0.18	0.4	0.4	75	0.06	0.9	20	4	20	3
Mass Casualty Event	Accident / disaster influx	300	65	10	30	0.22	0.5	0.3	65	0.15	1.3	30	4	20	3
ER Infrastructure Expansion	New ER wing added	200	45	20	45	0.12	0.3	0.45	85	0.11	1	20	4	35	4
Poor Hospital Layout / Wayfinding	Confusing corridors & signage	120	40	20	40	0.12	0.3	0.4	80	0.05	0.7	40	4	20	3
Preventive Care & Early Intervention	Strong primary healthcare system	60	30	15	50	0.06	0.2	0.5	90	0.12	1	20	4	20	3
Worst Case System Failure	Pandemic-level collapse	260	70	40	55	0.25	0.6	0.2	50	0.05	0.8	25	1	10	2

Results														
Time (Ticks)			Patients Saved			Patients Died (Outside ER)			Patients Died (In ER)			Mortality Percentage		
Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3
474	493	520	70	66	64	10	14	16	0	0	0	12.5	17.5	20
481	467	481	98	97	97	82	83	83	0	0	0	45.56	46.11	46.11
758	756	721	22	23	21	98	97	99	0	0	0	81.67	80.83	82.5
810	761	759	105	109	112	35	31	28	0	0	0	25	22.14	20
503	426	443	91	84	85	9	16	15	0	0	0	9	16	15
589	608	571	80	80	80	220	220	220	0	0	0	73.33	73.33	73.33
481	507	482	142	147	153	58	53	47	0	0	0	29	26.5	23.5
396	472	432	75	77	64	45	43	56	0	0	0	37.5	35.83	46.67
545	574	528	48	51	56	12	9	4	0	0	0	20	15	6.67
1047	1016	945	0	0	0	250	250	250	10	10	10	100	100	100

Table 2 with exploded view of tables given below for visibility (Intermittent treatment)

Scenario	Real World Analogy	Experimental Parameter Configuration													
		IP	IS	CT	TH	DR	ERDF	TR	ERSR	MS	SF	MTA	NERR	ERC	ERS
Normal Day Baseline ER	Typical weekday public hospital	80	35	20	40	0.08	0.25	0.45	85	0.1	1	20	3	20	3
Seasonal Flu Surge	Winter influenza increase	180	45	20	45	0.15	0.3	0.4	75	0.1	1	20	4	20	3
Resource Limited / Understaffed Hospital	Rural or low-income ER	120	45	30	35	0.14	0.5	0.25	60	0.07	0.8	20	1	10	2
Well Funded Urban Trauma Centre	Major metropolitan trauma hospital	140	50	15	50	0.07	0.2	0.6	90	0.12	1.2	25	4	30	4
Delayed Care / Low Health Awareness	Late presentation due to poor awareness	100	55	35	45	0.18	0.4	0.4	75	0.06	0.9	20	4	20	3
Mass Casualty Event	Accident / disaster influx	300	65	10	30	0.22	0.5	0.3	65	0.15	1.3	30	4	20	3
ER Infrastructure Expansion	New ER wing added	200	45	20	45	0.12	0.3	0.45	85	0.11	1	20	4	35	4
Poor Hospital Layout / Wayfinding	Confusing corridors & signage	120	40	20	40	0.12	0.3	0.4	80	0.05	0.7	40	4	20	3
Preventive Care & Early Intervention	Strong primary healthcare system	60	30	15	50	0.06	0.2	0.5	90	0.12	1	20	4	20	3
Worst Case System Failure	Pandemic-level collapse	260	70	40	55	0.25	0.6	0.2	50	0.05	0.8	25	1	10	2

Scenario	Results														
	Time (Ticks)			Patients Saved			Patients Died (Outside ER)			Patients Died (In ER)			Mortality Percentage		
	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3
Normal Day Baseline ER	836	878	854	64	65	69	16	15	11	0	0	0	20	18.75	13.75
Seasonal Flu Surge	1043	1058	1040	80	80	80	100	100	100	0	0	0	55.56	55.56	55.56
Resource Limited / Understaffed Hospital	1052	1093	1108	0	0	0	110	110	110	10	10	10	100	100	100
Well Funded Urban Trauma Centre	1022	956	1019	101	106	116	39	34	24	0	0	0	27.86	24.29	17.14
Delayed Care / Low Health Awareness	3368	4256	3524	77	79	72	23	21	28	0	0	0	23	21	28
Mass Casualty Event	452	466	448	0	0	0	220	220	220	80	80	80	100	100	100
ER Infrastructure Expansion	920	845	821	138	138	133	62	62	67	0	0	0	31	31	33.5
Poor Hospital Layout / Wayfinding	797	821	746	73	74	71	47	46	49	0	0	0	39.17	38.33	40.83
Preventive Care & Early Intervention	764	788	728	45	52	49	15	8	11	0	0	0	25	13.33	18.33
Worst Case System Failure	421	422	444	0	0	0	250	250	250	10	10	10	100	100	100

Scenario	Real World Analogy	Experimental Parameter Configuration													
		IP	IS	CT	TH	DR	ERDF	TR	ERSR	MS	SF	MTA	NERR	ERC	ERS
Normal Day Baseline ER	Typical weekday public hospital	80	35	20	40	0.08	0.25	0.45	85	0.1	1	20	3	20	3
Seasonal Flu Surge	Winter influenza increase	180	45	20	45	0.15	0.3	0.4	75	0.1	1	20	4	20	3
Resource Limited / Understaffed Hospital	Rural or low-income ER	120	45	30	35	0.14	0.5	0.25	60	0.07	0.8	20	1	10	2
Well Funded Urban Trauma Centre	Major metropolitan trauma hospital	140	50	15	50	0.07	0.2	0.6	90	0.12	1.2	25	4	30	4
Delayed Care / Low Health Awareness	Late presentation due to poor awareness	100	55	35	45	0.18	0.4	0.4	75	0.06	0.9	20	4	20	3
Mass Casualty Event	Accident / disaster influx	300	65	10	30	0.22	0.5	0.3	65	0.15	1.3	30	4	20	3
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Poor Hospital Layout / Wayfinding	Confusing corridors & signage	120	40	20	40	0.12	0.3	0.4	80	0.05	0.7	40	4	20	3
Preventive Care & Early Intervention	Strong primary healthcare system	60	30	15	50	0.06	0.2	0.5	90	0.12	1	20	4	20	3
Worst Case System Failure	Pandemic-level collapse	260	70	40	55	0.25	0.6	0.2	50	0.05	0.8	25	1	10	2

Results														
Time (Ticks)			Patients Saved			Patients Died (Outside ER)			Patients Died (In ER)			Mortality Percentage		
Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3
836	878	854	64	65	69	16	15	11	0	0	0	20	18.75	13.75
1043	1058	1040	80	80	80	100	100	100	0	0	0	55.56	55.56	55.56
1052	1093	1108	0	0	0	110	110	110	10	10	10	100	100	100
1022	956	1019	101	106	116	39	34	24	0	0	0	27.86	24.29	17.14
3368	4256	3524	77	79	72	23	21	28	0	0	0	23	21	28
452	466	448	0	0	0	220	220	220	80	80	80	100	100	100
920	845	821	138	138	133	62	62	67	0	0	0	31	31	33.5
797	821	746	73	74	71	47	46	49	0	0	0	39.17	38.33	40.83
764	788	728	45	52	49	15	8	11	0	0	0	25	13.33	18.33
421	422	444	0	0	0	250	250	250	10	10	10	100	100	100

<u>Legend to understand the sliders</u>

Disclaimer - Reported parameter ranges correspond to values explored in the experimental scenarios. All parameters remain adjustable within the NetLogo interface, and alternative configurations may be explored beyond those documented here.				
Abbreviation	**Parameter Name**	**Range Available**	**Description**	**Effect of Increasing / Decreasing Value**
IP	Initial Patients	10 - 300	Number of patient agents created at simulation start	↑ Increases system load and congestion; ↓ reduces demand and ER pressure
IS	Initial Severity	10 - 100	Baseline health assigned to patients at entry	↑ Patients enter healthier, delaying criticality; ↓ patients deteriorate sooner
CT	Critical Threshold	10 - 40	Health level below which patients begin seeking ER care	↑ Patients seek ER later, increasing risk of death outside ER; ↓ earlier ER demand
TH	Target Health	20 - 100	Health level required for recovery and discharge	↑ Longer treatment duration and ER occupancy; ↓ faster discharge and throughput
DR	Deterioration Rate	0.05 - 0.5	Amount of health lost per tick when untreated	↑ Faster health decline and higher mortality; ↓ slower deterioration
ERDF	ER Deterioration Factor	0 - 1	Multiplier applied to deterioration during ER treatment	↑ Less effective treatment (more decline in ER); ↓ greater stabilisation during treatment
TR	Treatment Rate	0.1 - 1	Health gained during a successful treatment event	↑ Faster recovery when treatment succeeds; ↓ slower health improvement
ERSR	ER Success Rate	0 - 100	Probability (%) that a treatment attempt is successful	↑ Higher probability of effective treatment; ↓ increased treatment failure
MS	Minimum Speed	0.05 - 0.3	Lower bound on patient movement speed	↑ Prevents immobility of critical patients; ↓ increases delays for low-health agents
SF	Speed Factor	0.5 - 2	Scales movement speed as a function of health	↑ Healthier patients move significantly faster; ↓ movement less sensitive to health
MTA	Maximum Turn Angle	5 - 50	Controls randomness in movement direction	↑ Less efficient navigation and increased delays; ↓ more directed movement
NERR	Number of ER Rooms	1 - 4	Number of spatially distinct ER treatment zones	↑ Distributes treatment load spatially; ↓ concentrates congestion
ERC	ER Capacity	1 - 50	Maximum patients treated simultaneously per ER room	↑ More patients treated simultaneously; ↓ increased admission blocking
ERS	ER Size	2 - 6	Physical patch area of each ER room	↑ Physical space for ER occupancy and queueing; ↓ increased spatial congestion

The parameter ranges reported reflect values explored within the experimental scenarios presented in this study. All parameters remain adjustable within the NetLogo interface, and alternative configurations may be explored beyond those documented here. The selected ranges were chosen to produce stable, interpretable, and non-trivial system behaviour rather than to exhaustively bound all possible outcomes.

6.2 Overview of Aggregate Outcomes

Across all scenarios, three dominant behavioural regimes emerge:

1. **Stable throughput**, characterised by low mortality and manageable congestion;
2. **Prolonged congestion**, marked by rising access-related deaths; and
3. **System collapse**, where demand overwhelms both access and treatment capacity.

Transitions between these regimes occur abruptly rather than gradually, indicating strong threshold effects. Small changes in demand, capacity, or access efficiency often produce disproportionately large increases in mortality and system duration.

Importantly, system clearance time does not correlate linearly with survival outcomes. In several scenarios, extended simulation runtimes reflect persistent congestion and delayed access rather than effective treatment throughput. Across most conditions, mortality is dominated by deaths occurring outside the emergency department, highlighting the primacy of access-related failure modes over treatment failure.

6.3 Comparative Scenario Outcomes (Quantitative Results)

Table 2 presents the averaged outcomes observed across all ten experimental scenarios using the final intermittent-treatment model. Metrics include total simulation time, number of patients saved, deaths occurring outside the ER, deaths occurring during ER treatment, and overall mortality percentage.

The table highlights pronounced non-linear effects. Scenarios characterised by early intervention, adequate capacity, and efficient spatial design consistently outperform those dominated by delayed care, congestion, or limited treatment availability. In contrast, combined stressors—such as increased demand coupled with reduced capacity—produce abrupt transitions to collapse, with near-total patient loss.

While individual run values vary due to stochastic movement and probabilistic treatment success, the relative ordering of scenarios and qualitative outcome patterns remain consistent across replications. This indicates that observed trends are structurally driven by model design rather than random variation.

Table 2 with exploded view of tables given below for visibility (Intermittent treatment)

Scenario	Real World Analogy	IP	IS	CT	TH	DR	ERDF	TR	ERSR	MS	SF	MTA	NERR	ERC	ERS
Normal Day Baseline ER	Typical weekday public hospital	80	35	20	40	0.08	0.25	0.45	85	0.1	1	20	3	20	3
Seasonal Flu Surge	Winter influenza increase	180	45	20	45	0.15	0.3	0.4	75	0.1	1	20	4	20	3
Resource Limited / Understaffed Hospital	Rural or low-income ER	120	45	30	35	0.14	0.5	0.25	60	0.07	0.8	20	1	10	2
Well Funded Urban Trauma Centre	Major metropolitan trauma hospital	140	50	15	50	0.07	0.2	0.6	90	0.12	1.2	25	4	30	4
Delayed Care / Low Health Awareness	Late presentation due to poor awareness	100	55	35	45	0.18	0.4	0.4	75	0.06	0.9	20	4	20	3
Mass Casualty Event	Accident / disaster influx	300	65	10	30	0.22	0.5	0.3	65	0.15	1.3	30	4	20	3
ER Infrastructure Expansion	New ER wing added	200	45	20	45	0.12	0.3	0.45	85	0.11	1	20	4	35	4
Poor Hospital Layout / Wayfinding	Confusing corridors & signage	120	40	20	40	0.12	0.3	0.4	80	0.05	0.7	40	4	20	3
Preventive Care & Early Intervention	Strong primary healthcare system	60	30	15	50	0.06	0.2	0.5	90	0.12	1	20	4	20	3
Worst Case System Failure	Pandemic-level collapse	260	70	40	55	0.25	0.6	0.2	50	0.05	0.8	25	1	10	2

Scenario	Time (Ticks) Iter 1	Iter 2	Iter 3	Patients Saved Iter 1	Iter 2	Iter 3	Patients Died (Outside ER) Iter 1	Iter 2	Iter 3	Patients Died (In ER) Iter 1	Iter 2	Iter 3	Mortality Percentage Iter 1	Iter 2	Iter 3
Normal Day Baseline ER	474	493	520	70	66	64	10	14	16	0	0	0	12.5	17.5	20
Seasonal Flu Surge	481	467	481	98	97	97	82	83	83	0	0	0	45.56	46.11	46.11
Resource Limited / Understaffed Hospital	758	756	721	22	23	21	98	97	99	0	0	0	81.67	80.83	82.5
Well Funded Urban Trauma Centre	810	761	759	105	109	112	35	31	28	0	0	0	25	22.14	20
Delayed Care / Low Health Awareness	503	426	443	91	84	85	9	16	15	0	0	0	9	16	15
Mass Casualty Event	589	608	571	80	80	80	220	220	220	0	0	0	73.33	73.33	73.33
ER Infrastructure Expansion	481	507	482	142	147	153	58	53	47	0	0	0	29	26.5	23.5
Poor Hospital Layout / Wayfinding	396	472	432	75	77	64	45	43	56	0	0	0	37.5	35.83	46.67
Preventive Care & Early Intervention	545	574	528	48	51	56	12	9	4	0	0	0	20	15	6.67
Worst Case System Failure	1047	1016	945	0	0	0	250	250	250	10	10	10	100	100	100

Scenario	Real World Analogy	Experimental Parameter Configuration													
		IP	IS	CT	TH	DR	ERDF	TR	ERSR	MS	SF	MTA	NERR	ERC	ERS
Normal Day Baseline ER	Typical weekday public hospital	80	35	20	40	0.08	0.25	0.45	85	0.1	1	20	3	20	3
Seasonal Flu Surge	Winter influenza increase	180	45	20	45	0.15	0.3	0.4	75	0.1	1	20	4	20	3
Resource Limited / Understaffed Hospital	Rural or low-income ER	120	45	30	35	0.14	0.5	0.25	60	0.07	0.8	20	1	10	2
Well Funded Urban Trauma Centre	Major metropolitan trauma hospital	140	50	15	50	0.07	0.2	0.6	90	0.12	1.2	25	4	30	4
Delayed Care / Low Health Awareness	Late presentation due to poor awareness	100	55	35	45	0.18	0.4	0.4	75	0.06	0.9	20	4	20	3
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Poor Hospital Layout / Wayfinding	Confusing corridors & signage	120	40	20	40	0.12	0.3	0.4	80	0.05	0.7	40	4	20	3
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Worst Case System Failure	Pandemic-level collapse	260	70	40	55	0.25	0.6	0.2	50	0.05	0.8	25	1	10	2

Results														
Time (Ticks)			Patients Saved			Patients Died (Outside ER)			Patients Died (In ER)			Mortality Percentage		
Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3	Iter 1	Iter 2	Iter 3
836	878	854	64	65	69	16	15	11	0	0	0	20	18.75	13.75
1043	1058	1040	80	80	80	100	100	100	0	0	0	55.56	55.56	55.56
1052	1093	1108	0	0	0	110	110	110	10	10	10	100	100	100
1022	956	1019	101	106	116	39	34	24	0	0	0	27.86	24.29	17.14
3368	4256	3524	77	79	72	23	21	28	0	0	0	23	21	28
452	466	448	0	0	0	220	220	220	80	80	80	100	100	100
920	845	821	138	138	133	62	62	67	0	0	0	31	31	33.5
797	821	746	73	74	71	47	46	49	0	0	0	39.17	38.33	40.83
764	788	728	45	52	49	15	8	11	0	0	0	25	13.33	18.33
421	422	444	0	0	0	250	250	250	10	10	10	100	100	100

6.4 Scenario-Based Time-Series Analysis

Time-series plots were analysed for all ten experimental scenarios to examine ER load, patient outcomes, and population-level health dynamics over time. Although each scenario represents a distinct real-world condition, results cluster into a small number of recurring behavioural regimes. To avoid unnecessary repetition, scenarios are therefore discussed in analytically coherent groups. Full scenario-wise plots are provided in below.

All figures are derived directly from the implemented NetLogo Web model and are included to illustrate correspondence between conceptual design, implementation logic, and observed system behaviour.

6.4.1 Stable and Resilient Operation Scenarios

(Scenarios 1, 4, and 9)

Under baseline operating conditions (Scenario 1), results from the intermittent-treatment model (Table 2) show ER load initially increasing as patients deteriorate and seek care, before stabilising and gradually declining once treatment throughput exceeds incoming demand. Mortality remains low and occurs almost entirely outside the ER, indicating brief access delays rather than treatment failure. Patient outcome plots show a steady increase in recoveries following early congestion, while health dynamics confirm that intermittent treatment reverses population-level deterioration once ER pressure stabilises.

The well-funded urban trauma centre scenario (Scenario 4) exhibits the same qualitative pattern but with faster stabilisation and earlier recovery onset. Increased ER capacity and improved treatment effectiveness suppress peak congestion and shorten the duration of system stress, resulting in lower mortality and faster clearance relative to the baseline.

The preventive care and early intervention scenario (Scenario 9) further reinforces this resilient regime by delaying the onset of critical deterioration, reducing peak ER demand, and accelerating recovery without requiring increased treatment capacity. Together, Scenarios 1, 4, and 9 demonstrate operating conditions under which the system maintains stable throughput and low mortality despite temporary congestion.

Plots


ER Pressure and Mortality
Number of Patients (Died)
Time (Ticks)
Avg ER Load
ER Deaths
Patient Outcome Over Time
Number of Patients
Time (Ticks)
Saved
Died (Non-ER)
Died in ER
Health Dynamics (Model Parameters)
Health Change per Tick
Time (Ticks)
Global Deterioration Rate
ER Deterioration Rate
Treatment Rate
Critical Threshold Rate
Target Health Rate
Average Patient Heath

Plots


ER Pressure and Mortality
Number of Patients (Died)
Time (Ticks)
Avg ER Load
ER Deaths
Patient Outcome Over Time
Number of Patients
Time (Ticks)
Saved
Died (Non-ER)
Died in ER
Health Dynamics (Model Parameters)
Health Change per Tick
Time (Ticks)
Global Deterioration Rate
ER Deterioration Rate
Treatment Rate
Critical Threshold Rate
Target Health Rate
Average Patient Heath

6.4.2 Congested and Capacity-Constrained Scenarios

(Scenarios 2, 3, 5, 7, and 8)

Moderately stressed scenarios in the intermittent-treatment model (Table 2) consistently exhibit prolonged congestion and elevated access-related mortality. In the seasonal flu surge scenario (Scenario

2), increased patient arrivals cause ER load to saturate for extended periods, with deaths occurring outside the ER rising sharply before recovery begins.

A similar pattern emerges in the resource-limited and understaffed hospital scenario (Scenario 3), where reduced treatment capacity accelerates saturation even in the absence of increased demand. In both scenarios, deaths during ER treatment remain limited, indicating that delayed access rather than treatment failure dominates system breakdown.

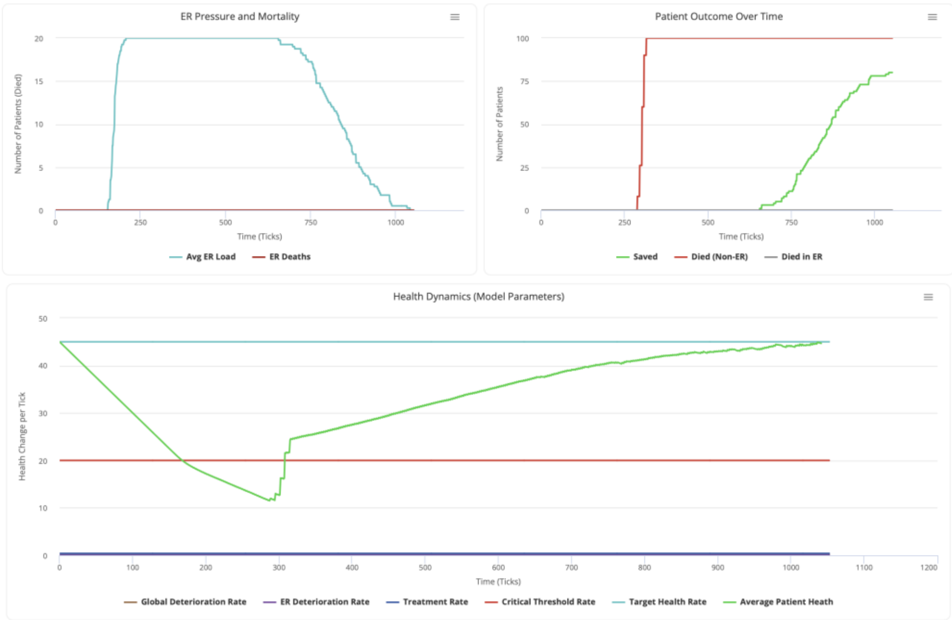
Delayed care due to low health awareness (Scenario 5) produces comparable outcomes despite unchanged capacity. Health dynamics plots show sustained periods during which average patient health remains below the critical threshold, reflecting deterioration prior to ER admission rather than ineffective treatment.

Infrastructure expansion without proportional staffing or treatment improvement (Scenario 7) delays but does not eliminate congestion. ER load peaks later than in strictly capacity-constrained scenarios, yet access-related mortality remains substantial, demonstrating diminishing returns from spatial expansion alone.

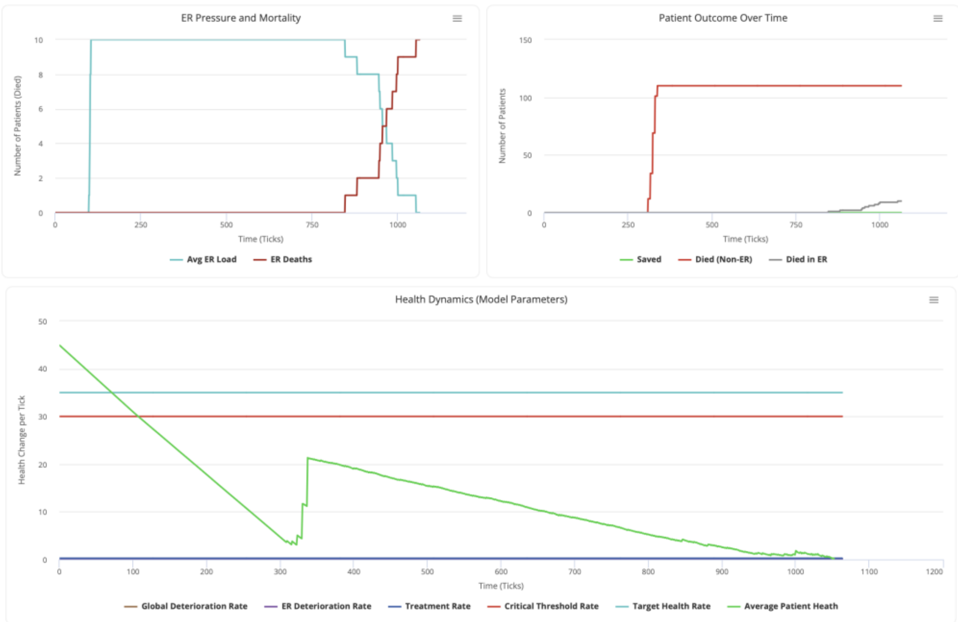
Poor hospital layout and inefficient wayfinding (Scenario 8) further exacerbate access delays. Despite unchanged treatment capacity, spatial inefficiency prolongs congestion and increases mortality, highlighting the critical role of environmental design alongside resource availability.

Collectively, Scenarios 2, 3, 5, 7, and 8 define a congested operating regime in which treatment remains effective once accessed, but delayed admission drives adverse outcomes.

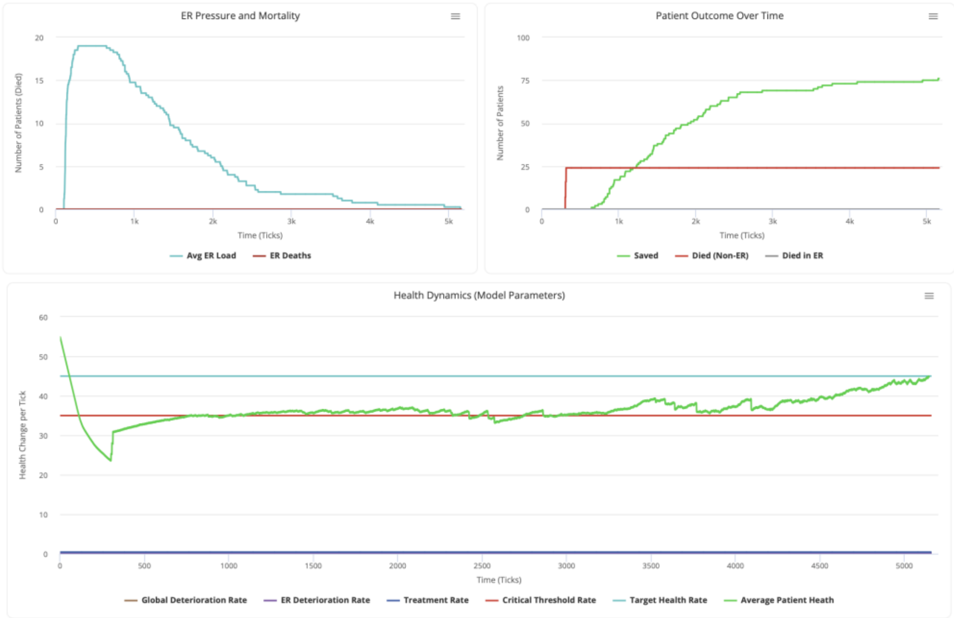
Plots


ER Pressure and Mortality
Number of Patients (Died)
Time (Ticks)
Avg ER Load
ER Deaths
Patient Outcome Over Time
Number of Patients
Time (Ticks)
Saved
Died (Non-ER)
Died in ER
Health Dynamics (Model Parameters)
Health Change per Tick
Time (Ticks)
Global Deterioration Rate
ER Deterioration Rate
Treatment Rate
Critical Threshold Rate
Target Health Rate
Average Patient Heath

Plots


ER Pressure and Mortality
Number of Patients (Died)
Time (Ticks)
Avg ER Load
ER Deaths
Patient Outcome Over Time
Number of Patients
Time (Ticks)
Saved
Died (Non-ER)
Died in ER
Health Dynamics (Model Parameters)
Health Change per Tick
Time (Ticks)
Global Deterioration Rate
ER Deterioration Rate
Treatment Rate
Critical Threshold Rate
Target Health Rate
Average Patient Heath

Plots


ER Pressure and Mortality
Number of Patients (Died)
Time (Ticks)
Avg ER Load
ER Deaths
Patient Outcome Over Time
Number of Patients
Time (Ticks)
Saved
Died (Non-ER)
Died in ER
Health Dynamics (Model Parameters)
Health Change per Tick
Time (Ticks)
Global Deterioration Rate
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Average Patient Heath

Plots


ER Pressure and Mortality
Number of Patients (Died)
Time (Ticks)
Avg ER Load
ER Deaths
Patient Outcome Over Time
Number of Patients
Time (Ticks)
Saved
Died (Non-ER)
Died in ER
Health Dynamics (Model Parameters)
Health Change per Tick
Time (Ticks)
Global Deterioration Rate
ER Deterioration Rate
Treatment Rate
Critical Threshold Rate
Target Health Rate
Average Patient Heath

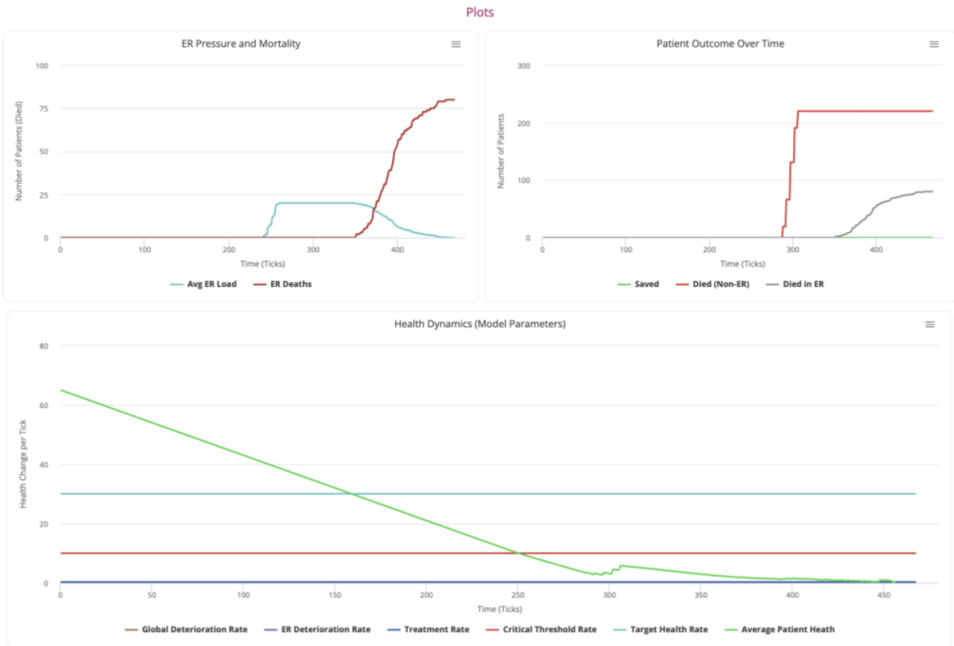
6.4.3 Collapse and Extreme Stress Scenarios

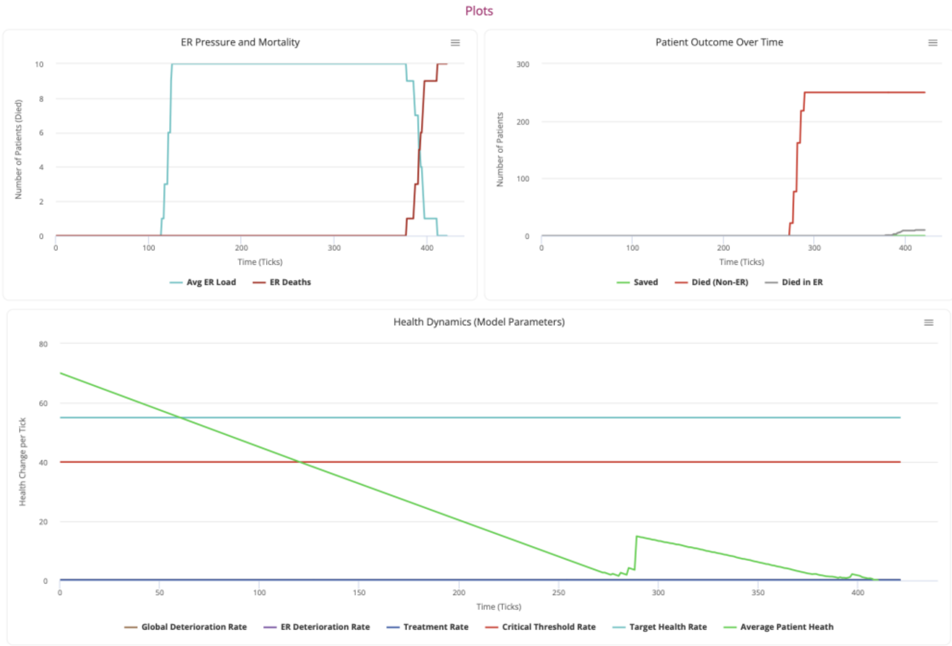
(Scenarios 6 and 10)

Under extreme stress conditions, the intermittent-treatment model (Table 2) exhibits abrupt transition into system collapse. In the mass casualty event scenario (Scenario 6), ER load reaches capacity rapidly and remains saturated, while mortality accelerates sharply. Recovery is both delayed and limited, indicating that treatment throughput is overwhelmed by incoming critical demand.

The worst-case system failure scenario (Scenario 10) demonstrates catastrophic collapse. ER capacity is saturated almost immediately, while deaths occurring outside the ER dominate outcomes. Deaths during ER treatment emerge only after sustained overload, confirming that treatment-related mortality is a secondary failure mode that signals deep systemic breakdown rather than routine operational stress.

In both scenarios, system clearance occurs primarily through mortality rather than recovery, marking a qualitative shift from congested operation to full systemic collapse.





6.5 Mortality Patterns and Failure Modes

Across the majority of scenarios, mortality occurs predominantly outside the emergency department rather than during treatment. This reflects the central role of delayed access, congestion, and spatial constraints in driving system failure. Deaths occurring during ER treatment remain rare under baseline and moderate stress conditions, emerging primarily under sustained overload.

Because only a fixed number of patients can be treated simultaneously, deaths during treatment are inherently bounded by capacity. In contrast, access-related mortality scales with patient demand and congestion. This reinforces the interpretation that interventions improving access efficiency and reducing system load are likely to be more effective than marginal improvements in treatment effectiveness alone.

6.6 Non-Linear Effects and Threshold Behaviour

Results across scenarios exhibit strong non-linear dynamics. Rather than degrading gradually, system performance often transitions abruptly from stability to collapse once critical thresholds are exceeded. These threshold effects arise from the interaction between continuous health deterioration, intermittent treatment, and finite treatment capacity.

Once ER load saturates, additional patients experience compounding access delays, producing rapid escalation in mortality. These dynamics highlight the suitability of agent-based modelling for capturing emergent behaviour and phase-transition-like effects that are difficult to represent using aggregate or equilibrium-based analytical approaches.

6.7 Cross-Scenario Synthesis

Comparative analysis reveals three dominant emergency department operating regimes:
(1) **Stable throughput**, characterised by short-lived congestion and low mortality;
(2) **Congested operation**, marked by prolonged saturation and rising access-related deaths; and
(3) **System collapse**, where demand overwhelms both access and treatment capacity.

Preventive care, early intervention, and adequate capacity consistently shift the system toward stable regimes, while compounded stressors trigger abrupt collapse. These findings underscore the importance of upstream healthcare investment, spatial efficiency, and resilience-focused system design.

6.8 Summary of Results

Overall, the results demonstrate that emergency department outcomes are driven more strongly by access constraints and congestion than by treatment effectiveness alone. Mortality emerges primarily from delayed access to care, with deaths during treatment acting as indicators of extreme systemic overload. The model illustrates how complex service systems can appear stable until critical thresholds are crossed, after which failure occurs rapidly and non-linearly.

7. Evaluation and Limitations

The model successfully demonstrates how emergency department outcomes emerge from individual behaviour, spatial constraints, and capacity limitations. By distinguishing deaths occurring outside the emergency department from those occurring during treatment, the model provides insight into dominant failure modes under varying stress conditions.

The revised treatment dynamics strengthen the model by ensuring that emergency department mortality is neither trivial nor ubiquitous. Instead, deaths occurring during treatment are bounded by capacity and emerge primarily under sustained overload, reinforcing their interpretation as indicators of systemic collapse rather than routine operation.

While the model includes deliberate simplifications, these abstractions improve interpretability by isolating the core mechanisms driving system behaviour. Increased realism would not necessarily improve explanatory power and could obscure emergent dynamics central to the study's objectives.

7.1 Evaluation of Model Performance

The agent-based model developed in this project successfully demonstrates how emergency department outcomes emerge from the interaction of individual patient behaviour, spatial constraints, and limited treatment capacity. Across all scenarios, the model produces coherent, interpretable, and non-trivial system dynamics, including congestion, delayed access to care, and non-linear increases in mortality under stress.

A key strength of the model is its ability to differentiate between access-related failures and treatment-related failures. By explicitly tracking deaths occurring outside the emergency department separately from those occurring during treatment, the model reveals where and how system breakdown occurs. This distinction is particularly valuable in high-load scenarios, where interventions targeting only one failure mode may be insufficient.

Across most scenarios, mortality predominantly occurs outside the emergency department rather than during treatment. This reflects the model's emphasis on delayed access and capacity bottlenecks, where patients frequently deteriorate to fatal levels before treatment can begin. Deaths occurring within the emergency department are observed primarily under extreme collapse conditions, indicating that treatment failure is a secondary failure mode relative to access failure.

The model also demonstrates strong internal consistency. Variations in outcomes across scenarios align with intuitive expectations while still revealing threshold effects. Moderate increases in patient demand result in gradual performance degradation, whereas combined stressors produce abrupt system collapse. These behaviours suggest that the model captures essential features of emergency department dynamics rather than artefacts of parameter choice.

The concentration of mortality outside the emergency department is therefore interpreted as a model insight rather than a limitation, reflecting the dominance of access failure over treatment failure in capacity-constrained systems. While mortality outcomes are simplified to discrete state transitions rather than continuous clinical trajectories, this abstraction supports computational tractability and highlights emergent system behaviour.

7.2 Evaluation of Conceptual Parallelism

Although explicit thread-level parallelism is not implemented due to the constraints of the NetLogo Web environment, the model is inherently parallel at the agent level. Each patient agent updates its state independently during each simulation tick, with interaction occurring only through shared environmental constraints such as emergency department capacity.

This design mirrors the conceptual structure of parallel systems in which independent tasks are executed concurrently and synchronised at discrete intervals. As such, the model provides a valid demonstration of parallel computing principles within an agent-based framework. The independence of agent updates suggests that the model could scale efficiently in environments supporting explicit multi-threading, with synchronisation required only for emergency department admission and discharge events.

However, because execution remains sequential at the platform level, performance characteristics such as scheduling overhead and contention cannot be empirically evaluated within the current implementation.

7.2.1 Validity Considerations

The model demonstrates strong internal validity, as agent rules behave consistently with their intended design and produce stable outcomes across repeated simulation runs. State transitions, capacity checks, and outcome counters operate without logical contradictions, ensuring that observed behaviours arise from model structure rather than execution artefacts.

Construct validity is supported by the alignment between model abstractions and the real-world phenomena under investigation. Key mechanisms such as delayed access to care, spatial congestion, capacity-constrained treatment, and overload-driven failure are represented explicitly and interact in a manner consistent with emergency department dynamics.

External validity is intentionally limited. The model is not designed to generate quantitative predictions for real emergency departments, but rather to provide qualitative insight into system behaviour, threshold effects, and failure modes under stress. These validity boundaries are appropriate given the exploratory objectives of the study.

7.2.2 Execution Environment and Observed System-Level Parallelism

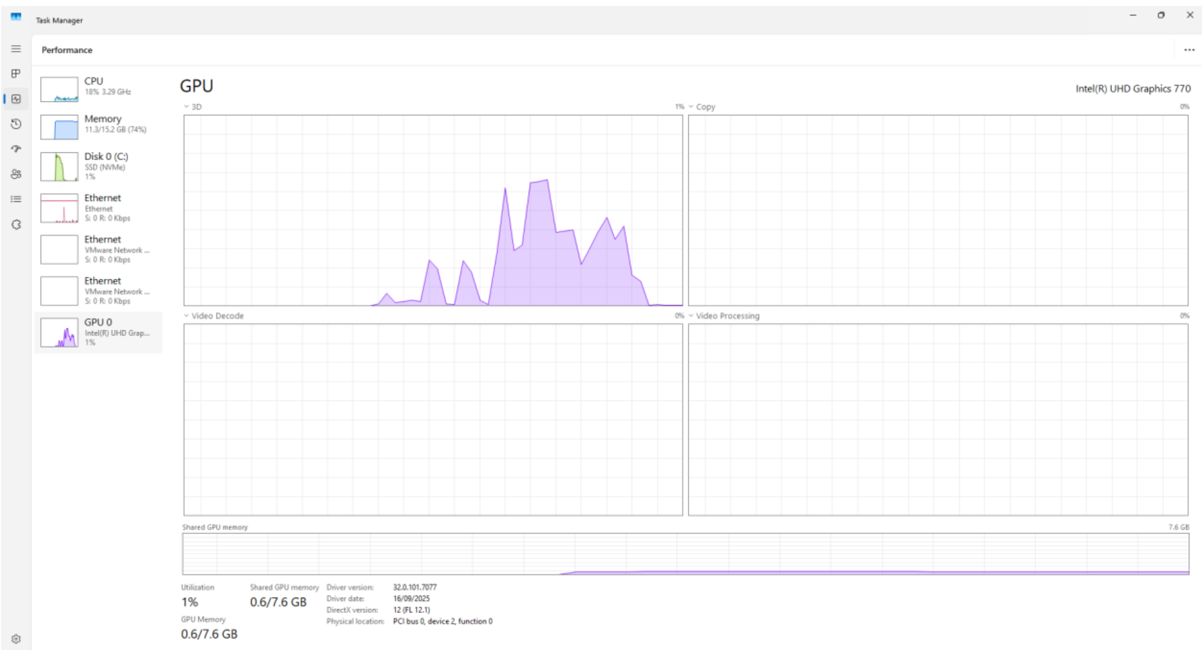
Although explicit thread-level multi-threading is not implemented within the NetLogo Web model itself, the simulation executes within a modern browser environment that inherently supports parallel execution across CPU cores and GPU resources.

During model execution, system-level monitoring using the operating system task manager indicates concurrent utilisation of multiple processing units, including CPU scheduling across cores and GPU acceleration for rendering and visual updates. This behaviour reflects the parallel execution model of web-based simulation environments, where computation, rendering, and event handling occur concurrently within the browser runtime.

It is important to note that this observed parallelism arises from the execution environment rather than from explicit user-defined threads in the model code. The agent-based logic remains conceptually parallel, with each patient agent evaluating health, movement, and treatment conditions independently within each simulation tick. These independent computations are then synchronised at discrete time steps, consistent with the synchronous update semantics of NetLogo.

The observed system-level parallel activity therefore supports the interpretation of the model as **conceptually parallel**, even though low-level thread management is abstracted away by the NetLogo Web platform and browser runtime. This distinction aligns with the objectives of the coursework, which emphasise parallel thinking and agent independence rather than explicit thread programming.

Although explicit thread-level multithreading is not implemented within the NetLogo Web model, execution occurs within a modern browser runtime that supports concurrent processing across CPU and GPU resources. System-level monitoring during simulation execution indicates parallel utilisation of computational and rendering resources. This parallel activity arises from the execution environment rather than from user-defined threads. Agent-level logic remains conceptually parallel, with each agent updating independently within a synchronous time step. This distinction aligns with the coursework objective of demonstrating parallel thinking through agent independence rather than low-level thread management.



System-level monitoring during simulation execution showing concurrent CPU and GPU activity within the browser runtime environment.

7.3 Model Limitations

Despite its strengths, the model includes several simplifications that limit its direct applicability to real-world emergency departments.

7.3.1 No Spontaneous Recovery Outside the Emergency Department

Patients do not recover spontaneously outside the emergency department; untreated deterioration is assumed to be monotonic. This simplification ensures that access delays have irreversible consequences and prevents recovery without clinical intervention. While this limits clinical realism, it strengthens interpretability by isolating access-related failure as a dominant system mechanism.

7.3.2 Absence of Staff or Resource Agents

The model does not explicitly represent medical staff, equipment, or triage personnel. Emergency department capacity is treated as a static parameter rather than an emergent property of staff availability or skill level. While this abstraction simplifies the model, it limits the ability to explore staffing strategies or shift-based resource allocation.

7.3.3 Simplified Health Dynamics

Health deterioration and recovery are modelled using fixed rates and probabilistic treatment success. In reality, patient trajectories are influenced by numerous factors such as age, comorbidities, and prioritisation. The absence of patient subtypes limits the realism of individual outcomes.

7.3.4 Lack of Triage and Prioritisation

Patients are treated on a first-available basis without explicit triage rules. Real emergency departments prioritise patients based on severity, which can significantly alter mortality patterns. The absence of triage may contribute to the dominance of access-related mortality observed in high-load scenarios, as patients are not selectively admitted based on criticality.

7.3.5 Static Spatial Layout

The hospital layout is fixed within each scenario. While this supports controlled experimentation, it limits exploration of adaptive routing or dynamic spatial reconfiguration in response to congestion.

7.3.6 Limited Statistical Analysis

Each scenario is replicated three times to account for stochastic variation. While sufficient for identifying qualitative trends, a larger number of runs would be required for rigorous statistical inference. Such analysis was beyond the scope of this coursework.

7.4 Implications of Limitations

These limitations do not undermine the core insights of the model. Simplifications were chosen deliberately to focus on system-level dynamics rather than clinical detail. The model's ability to reproduce congestion, non-linear collapse, and differential failure modes suggests that the observed behaviours are robust within the defined abstraction scope.

By isolating mechanisms such as capacity constraints and spatial delays, the model provides explanatory clarity that might be obscured in more detailed simulations.

7.5 Potential Extensions and Improvements

Several extensions could enhance realism and analytical power, including:

- introducing staff agents with shift schedules and fatigue effects,
- implementing triage rules based on patient severity,
- modelling heterogeneous patient types with distinct deterioration profiles,
- allowing dynamic reallocation of emergency department capacity,
- conducting large-scale statistical experiments to quantify uncertainty.

These extensions would enable exploration of additional policy and operational questions while retaining the agent-based foundation of the model.

7.6 Summary

Overall, the model achieves its primary objectives: to explore how emergency department outcomes emerge from individual behaviour and system constraints, and to demonstrate the applicability of agent-based modelling and conceptual parallelism to complex healthcare systems. While simplified, the model provides meaningful insights into system resilience, dominant failure modes, and the potential impact of different intervention strategies.

8. Reflection on Learning

This project provided a significant opportunity to deepen my understanding of agent-based modelling, parallel computing concepts, and the challenges involved in representing complex real-world systems through simulation. While the initial goal was to construct a functional model of emergency department dynamics, the development process ultimately reshaped how I think about modelling, system behaviour, and computational abstraction.

One of the most important lessons learned was the distinction between individual-level logic and system-level outcomes. Early in the development process, it was tempting to focus on tuning parameters to achieve desirable aggregate results. However, as the model evolved, it became clear that meaningful outcomes must arise organically from local interactions rather than being imposed globally. Small changes to individual deterioration rates or movement rules often produced disproportionately large changes in overall mortality, reinforcing the importance of emergence as a defining characteristic of complex systems.

The project also deepened my appreciation for non-linear dynamics and threshold effects. Several scenarios demonstrated that system collapse did not occur gradually, but instead through abrupt transitions once critical capacity limits were exceeded. Observing these tipping points challenged assumptions about linear scalability and highlighted why emergency departments can appear stable for extended periods before suddenly becoming overwhelmed. This insight has relevance beyond healthcare modelling, extending to domains such as traffic systems, supply chains, and networked infrastructures.

From a technical perspective, the project clarified the meaning of parallelism within agent-based models. Although explicit thread-level multi-threading was not implemented due to platform constraints, the simulation nevertheless functioned as a conceptually parallel system. Each agent updated its state independently within a synchronous time step, mirroring the simultaneous operation of real-world entities.

Understanding this distinction helped me recognise that parallel computing is not solely about low-level thread management, but also about structuring problems in a way that enables independent execution and scalable design. This shift in perspective strengthened my understanding of how agent-based models naturally align with parallel computing principles, even within environments that execute sequentially.

Developing the model also highlighted the importance of intentional simplification. Real emergency departments involve numerous interacting components, including staff roles, triage protocols, and medical specialisations. Attempting to model all of these aspects would have obscured the core mechanisms under investigation. By deliberately abstracting away certain details, the model was able to focus on capacity constraints, spatial access, and deterioration dynamics, resulting in clearer and more interpretable insights. This reinforced the idea that an effective model is not one that includes everything, but one that includes the right elements for the research question being addressed.

Another key learning outcome was the value of scenario-based experimentation. Rather than treating the model as a single predictive tool, using multiple contrasting scenarios revealed patterns that would not have been visible in isolation. Comparing stable, stressed, and collapse regimes emphasised how system behaviour depends on context, and how interventions effective under one set of conditions may fail under another. This approach strengthened both my analytical skills and my understanding of experimental design in simulation-based research.

Finally, the process of documenting the model in a structured technical report significantly improved my ability to communicate complex ideas clearly and rigorously. Translating code into academic explanation required careful justification of design choices and explicit articulation of assumptions. This reflection reinforced the importance of clear communication in computational modelling, where the value of a model depends not only on its correctness, but also on how effectively its logic and implications are conveyed to others.

Overall, this project has strengthened my confidence in applying agent-based modelling to complex systems and has provided valuable insight into the interplay between individual behaviour, system constraints, and emergent outcomes. The skills and perspectives developed through this work will be directly applicable to future projects involving simulation, parallel systems, and complex real-world problem solving.

This project reinforced the importance of modelling discipline, intentional abstraction, and careful interpretation of emergent behaviour—skills that extend beyond simulation and are applicable to a wide range of complex computational systems.

9. Conclusion

This project demonstrates how agent-based modelling can be used to explore emergency department dynamics under capacity constraints and time-critical conditions. By representing patients as autonomous agents whose health, movement, and access to care evolve over time, the model captures emergent system behaviour including congestion, non-linear increases in mortality, and abrupt transitions from stable operation to systemic collapse.

The results show that access efficiency and capacity constraints play a dominant role in determining patient outcomes. Across multiple scenarios, mortality was driven primarily by delayed access to care rather than treatment failure, highlighting access-related bottlenecks as a critical vulnerability in capacity-constrained systems. Scenarios incorporating early intervention or improved initial health conditions demonstrated substantially greater resilience, reinforcing the importance of upstream measures in mitigating downstream system pressure.

From a methodological perspective, the model illustrates the strength of agent-based approaches for studying complex service systems in which global outcomes emerge from local interactions and shared constraints. The use of scenario-based experimentation enabled systematic comparison across operating regimes while preserving interpretability, and the distinction between access-related and treatment-related failure modes provided insight into how different forms of system stress manifest.

Although the model is intentionally abstract and not intended for direct clinical prediction, it provides meaningful qualitative insight into how emergency departments and similar service systems behave near capacity limits. More broadly, the findings underscore how complex systems can appear stable under moderate load yet fail abruptly when critical thresholds are exceeded. These insights extend beyond emergency healthcare and are applicable to other domains characterised by congestion, limited resources, and time-sensitive demand.